AI_ASSISTED ASSIGNMENT 14.3

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BATCH:46

Task 1: AI-Assisted Personal Portfolio Website

Scenario

A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure with sections: ○ Home ○ About ○ Projects ○ Contact
2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.
3. Customize AI-generated CSS to add:
   - Hover effects on project cards ○ Smooth scrolling navigation

Expected Outcome

- Clean and well-structured portfolio website
- Responsive layout for desktop and mobile screens
- Interactive project cards

CODE:

index.html styles.css script.js

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Your Name | Portfolio</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <nav>
    <ul>
      <li><a href="#home">Home</a></li>
      <li><a href="#about">About</a></li>
      <li><a href="#projects">Projects</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>
  <section id="home">
    <h1>Welcome to My Portfolio</h1>
    <p>Short introduction or tagline.</p>
  </section>
  <section id="about">
    <h2>About Me</h2>
    <p>Insert your biography, education, and skill summary here.</p>
  </section>
  <section id="projects">
    <h2>Projects</h2>
    <div class="projects-grid">
      <div class="project-card">
        <h3>Project Title</h3>
        <p>Project description.</p>
        <a href="#">View More</a>
      </div>
      <!-- Duplicate project-card as needed -->
    </div>
  </section>
  <section id="contact">
    <h2>Contact</h2>
    <form>
      <label for="email">Email:</label>
      <input type="email" id="email" required>
      <label for="message">Message:</label>
```

```html
        <a href="#">View More</a>
      </div>
      <!-- Duplicate project-card as needed -->
    </div>
  </section>
  <section id="contact">
    <h2>Contact</h2>
    <form>
      <label for="email">Email:</label>
      <input type="email" id="email" required>
      <label for="message">Message:</label>
      <textarea id="message" required></textarea>
      <button type="submit">Send</button>
    </form>
  </section>
</body>
</html>
```

OUTPUT:

Task 2: AI-Generated Restaurant Landing Page

Scenario

A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.

Tasks

1. Use Copilot to create a navigation bar with links:
   - Home
   - Menu
   - Gallery
   - Contact
2. Generate a menu section styled with CSS cards.
3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.
4. Customize animations or transitions for better user experience. **Expected Outcome**

- Fully functional restaurant landing page
- Responsive navigation bar
- Working image slider with smooth transitions

CODE:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Delizioso Restaurant</title>
  <link rel="stylesheet" href="styles.css"/>
</head>
<body>
  <!-- Navigation Bar -->
  <nav class="navbar">
    <div class="nav-container">
      <h1 class="logo">Delizioso</h1>
      <ul class="nav-links">
        <li><a href="#home">Home</a></li>
        <li><a href="#menu">Menu</a></li>
        <li><a href="#gallery">Gallery</a></li>
        <li><a href="#contact">Contact</a></li>
      </ul>
    </div>
  </nav>

  <section id="home" class="hero">
    <h2>Welcome to Delizioso</h2>
    <p>Experience authentic cuisine with a modern twist.</p>
  </section>

  <!-- Menu Section -->
  <section id="menu">
    <h2>Our Menu</h2>
    <div class="menu-cards">
      <div class="menu-card">
        <img src="https://profusioncurry.com/wp-content/uploads/2022/10/Pasta-Fresca-recipe-made-at-home.jpg"
        <h3>Pasta Fresca</h3>
        <p>Homemade noodles with fresh tomato sauce.</p>
        <span>$14</span>
      </div>
      <div class="menu-card">
        <img src="https://www.simplyrecipes.com/thmb/KRw_r32s4gQeOX-d07NWY1OlOFk=/1500x0/filters:no_upscale()
        <h3>Classic Pizza</h3>
        <p>Stone-baked with your choice of toppings.</p>
```

```html
        <p>Stone-baked with your choice of toppings.</p>
        <span>$12</span>
      </div>
      <div class="menu-card">
        <img src="https://www.recipetineats.com/tachyon/2021/08/Garden-Salad_48.jpg?resize=900%2C1260&zoom=0.
        <h3>Garden Salad</h3>
        <p>Crisp greens with house vinaigrette.</p>
        <span>$8</span>
      </div>
    </div>
  </section>

  <!-- Gallery Section -->
  <section id="gallery">
    <h2>Gallery</h2>
    <div class="slider">
      <div class="slides">
        <img src="slide1.jpg" alt="Interior" class="slide active"/>
        <img src="slide2.jpg" alt="Dish" class="slide"/>
        <img src="slide3.jpg" alt="Family" class="slide"/>
      </div>
      <button class="prev">&#10094;</button>
      <button class="next">&#10095;</button>
    </div>
  </section>

  <!-- Contact Section -->
  <section id="contact">
    <h2>Contact Us</h2>
    <form>
      <input type="text" placeholder="Your Name" required/>
      <input type="email" placeholder="Your Email" required/>
      <textarea placeholder="Message..." required></textarea>
      <button type="submit">Send</button>
    </form>
  </section>

  <script src="script.js"></script>
</body>
</html>
```

```css
/* General Reset */
body, h1, h2, h3, p, ul, li { margin:0; padding:0;}
body { font-family: 'Segoe UI', Arial, sans-serif; background: #fff; color: #333; }

/* Navbar */
.navbar {
  background: #af3e2b;
  color: #fff;
  position: sticky; top: 0; z-index: 100;
  box-shadow: 0 2px 8px rgba(0,0,0,0.07);
}
.nav-container {
  display: flex; justify-content: space-between; align-items: center;
  max-width: 1080px; margin: auto; padding: 1rem;
}
.logo { font-size: 1.7rem; letter-spacing: 2px; }
.nav-links { display: flex; list-style: none; }
.nav-links li {margin-left: 2rem;}
.nav-links a {
  color: #fff; text-decoration: none; font-weight: bold; font-size: 1.1rem;
  position: relative; transition: color 0.2s;
}
.nav-links a:after {
  content: ""; display: block; width: 0%; height: 2px;
  background: #fff; transition: width 0.3s;
  position: absolute; left: 0; bottom: -4px;
}
.nav-links a:hover:after { width: 100%; }
.nav-links a:hover { color: #ffe2cd; }

/* Hero */
.hero { text-align: center; padding: 80px 20px 40px 20px; background: #ffe2cd;}
.hero h2 { font-size: 2.3rem; margin-bottom: 0.4em; color: #af3e2b; }

/* Menu Section */
#menu { padding:60px 20px; background: #fff6f0;}
#menu h2 { text-align: center; font-size: 2rem; color: #bf452b;}
.menu-cards {
  display: flex; gap: 2rem; flex-wrap: wrap;
  justify-content: center; margin-top: 2rem;
}
```

STYLE CSS:

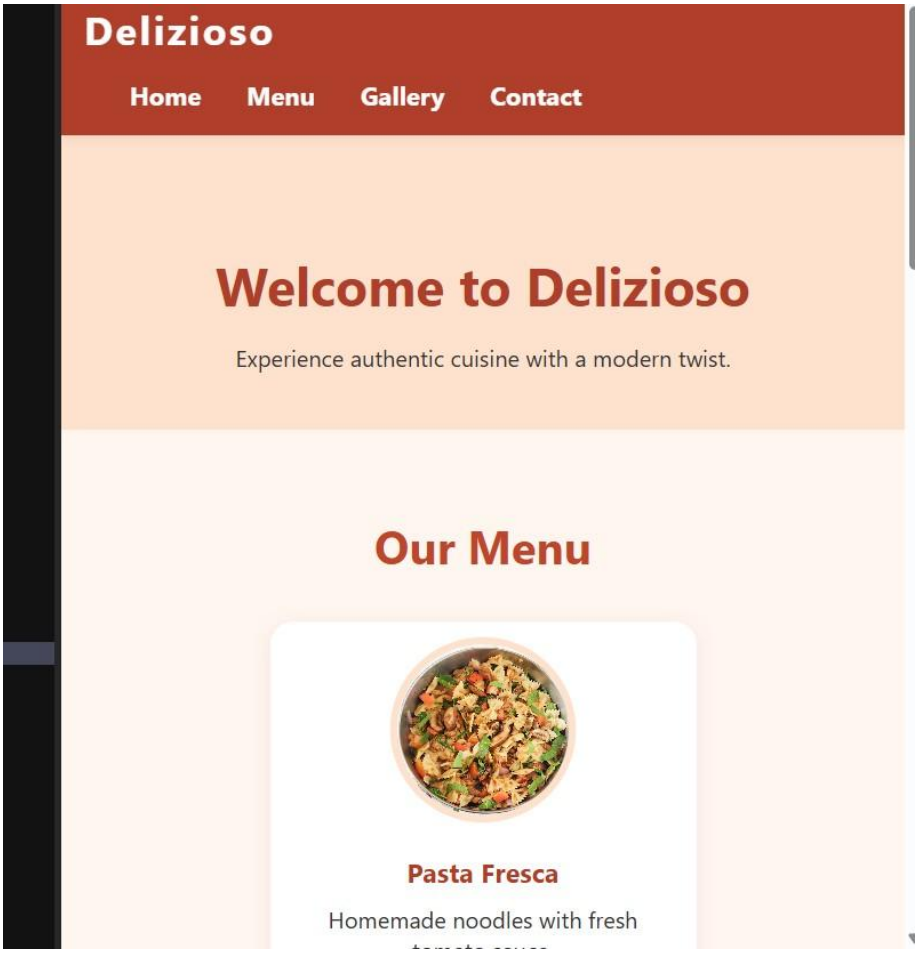
```c
.menu-card {
  background: #fff; border-radius: 1rem; box-shadow: 0 3px 14px rgba(175,62,43,0.10);
  width:260px; padding: 1em 1.2em 1.5em 1.2em; text-align: center;
  transition: transform 0.22s, box-shadow 0.21s;
  cursor: pointer;
  position: relative;
}
.menu-card:hover {
  transform: translateY(-8px) scale(1.045);
  box-shadow: 0 10px 30px rgba(175,62,43,0.17);
}
.menu-card img {
  width: 120px; height: 120px; border-radius: 50%; object-fit: cover;
  margin-bottom: 1em; box-shadow:0 0 0 5px #ffe2cd;
}
.menu-card h3 { margin:0.4em 0 0.5em 0; color:#b03e26;}
.menu-card span {display: block; font-weight: 600; color:#af3e2b; margin-top:0.7em;}

/* Gallery Section & Slider */
#gallery { padding: 60px 20px;}
#gallery h2 { text-align:center; color:#bf452b;}
.slider {
  position: relative; max-width: 420px; margin: 2rem auto 0 auto; overflow: hidden; border-radius:10px; backgr
  box-shadow:0 2px 20px #dda17b29;
}
.slides { display: flex; transition: transform 0.7s cubic-bezier(.52,.17,.21,.83);}
.slide {
  min-width: 100%; opacity:0; transition: opacity 0.7s;
  height: 270px; object-fit: cover;Menu
Gallery Images:



}
.slide.active { opacity:1; }
.prev, .next {
  position: absolute; top: 50%; transform: translateY(-50%);
  background: #af3e2b; color:#fff; border:none; font-size:1.8rem;
  border-radius: 50%; padding: 0.5em 0.8em; cursor:pointer;
  opacity:0.7; transition: background 0.2s, opacity 0.2s;
```

```c
}
.slide.active { opacity:1; }
.prev, .next {
  position: absolute; top: 50%; transform: translateY(-50%);
  background: #af3e2b; color:#fff; border:none; font-size:1.8rem;
  border-radius: 50%; padding: 0.5em 0.8em; cursor:pointer;
  opacity:0.7; transition: background 0.2s, opacity 0.2s;
}
.prev:hover, .next:hover { background:#d65511; opacity:1; }
.prev { left: 10px; }
.next { right: 10px; }

/* Contact Form */
#contact { padding:60px 20px; background: #ffe2cd;}
#contact h2 {text-align: center; color: #bf452b;}
form {
  max-width: 420px; margin:1.2em auto 0 auto; display: flex; flex-direction: column; gap: 1rem;
  box-shadow: 0 0 18px #b9612940; padding:1.5em 2em; border-radius: 11px; background:#fff;
}
input, textarea { font: inherit; padding: 0.7em; border-radius: 7px; border: 1px solid #bf452b48; outline: n
button[type="submit"] {
  background: #af3e2b; color: #fff; border: none; padding: 0.85em;
  border-radius: 6px; cursor: pointer; font-weight:600; font-size:1rem;
  transition: background 0.18s;
}
button[type="submit"]:hover { background: #c95c18; }

/* Responsive */
@media (max-width: 800px) {
  .menu-cards { flex-direction: column; align-items: center; }
  .nav-container { flex-direction: column; align-items: flex-start;}
  .logo { margin-bottom: 0.6em;}
}
@media (max-width: 480px) {
  .slider, form { max-width:98vw;}
}
```

SCRIPT JS:

```javascript
// Image slider logic
const slides = document.querySelectorAll('.slide');
const prevBtn = document.querySelector('.prev');
const nextBtn = document.querySelector('.next');
let currentIndex = 0;

function showSlide(index) {
  slides.forEach((slide, i) => {
    slide.classList.toggle('active', i === index);
  });
}

function nextSlide() {
  currentIndex = (currentIndex + 1) % slides.length;
  showSlide(currentIndex);
}

function prevSlide() {
  currentIndex = (currentIndex - 1 + slides.length) % slides.length;
  showSlide(currentIndex);
}

nextBtn.addEventListener('click', nextSlide);
prevBtn.addEventListener('click', prevSlide);

// Auto-advance every 5s
setInterval(nextSlide, 5000);
```

OUTPUT:



Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields: ○ Name ○ Email ○ Phone Number ○ Department ○ Event Selection
2. Use AI assistance to style the form using CSS for readability and accessibility.
3. Implement JavaScript validation using Copilot:
   - Email format validation ○ Phone number length validation
   - Required field checks

Expected Outcome

- User-friendly and visually appealing form
- Real-time validation messages
- Prevention of invalid submissions

CODE :

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
  <meta name="viewport" content="width=device-width,initial-scale=1.0"/>
  <title>SRU Tech Fest | Registration</title>
  <link rel="stylesheet" href="form.css"/>
</head>
<body>
  <main>
    <h1>SR University Technical Fest Registration</h1>
    <form id="regForm" novalidate>
      <div class="form-group">
        <label for="name">Name<span aria-label="required">*</span>:</label>
        <input type="text" id="name" name="name" required/>
        <small class="error"></small>
      </div>
      <div class="form-group">
        <label for="email">Email<span aria-label="required">*</span>:</label>
        <input type="email" id="email" name="email" required/>
        <small class="error"></small>
      </div>
      <div class="form-group">
        <label for="phone">Phone Number<span aria-label="required">*</span>:</label>
        <input type="tel" id="phone" name="phone" maxlength="10" pattern="[0-9]{10}" required/>
        <small class="error"></small>
      </div>
      <div class="form-group">
        <label for="dept">Department<span aria-label="required">*</span>:</label>
        <select id="dept" name="dept" required>
          <option value="">--Select Department--</option>
          <option value="cse">CSE</option>
          <option value="ece">ECE</option>
          <option value="eee">EEE</option>
          <option value="mech">MECH</option>
          <option value="civil">CIVIL</option>
        </select>
        <small class="error"></small>
      </div>
      <div class="form-group">
        <label for="event">Event Selection<span aria-label="required">*</span>:</label>
        <select id="event" name="event" required>
```


```html
      <div class="form-group">
        <label for="event">Event Selection<span aria-label="required">*</span>:</label>
        <select id="event" name="event" required>
```

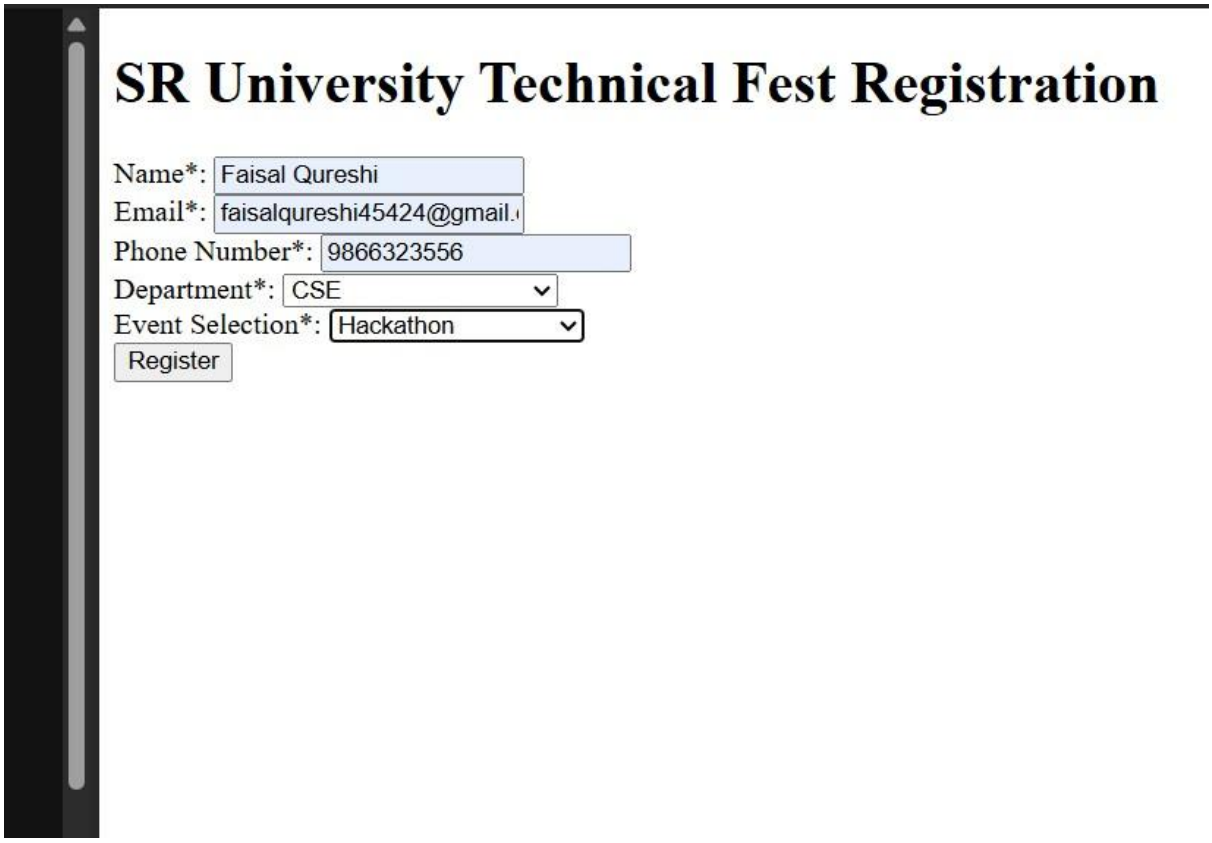

```html
          <option value="">--Select Event--</option>
          <option value="hackathon">Hackathon</option>
          <option value="poster">Poster Presentation</option>
          <option value="quiz">Tech Quiz</option>
          <option value="coding">Coding Contest</option>
        </select>
        <small class="error"></small>
      </div>
      <button type="submit">Register</button>
    </form>
    <div id="result" role="alert"></div>
  </main>
  <script src="form.js"></script>
</body>
</html>
```

```css
body {
  background: #f2f2fa;
  font-family: 'Segoe UI', Arial, sans-serif;
  margin: 0;
}
main {
  max-width: 420px;
  background: #fff;
  margin: 3.5em auto 0 auto;
  padding: 2em 2.5em;
  border-radius: 1.2em;
  box-shadow: 0 3px 24px #8f8beb33;
}
h1 {
  font-size: 1.7em;
  color: #5318a6;
  text-align: center;
  margin-bottom: 1.6em;
}
.form-group {
  margin-bottom: 1.35em;
  display: flex;
  flex-direction: column;
}
label {
  font-weight: 600;
  margin-bottom: 0.35em;
}
input, select {
  padding: 0.7em;
  border: 1.2px solid #bebfe5;
  border-radius: 6px;
  font-size: 1em;
  background: #fafaff;
  transition: border 0.19s;
}
input:focus, select:focus {
  border-color: #5318a688;
  outline: none;
}
button[type="submit"] {
  margin-top: 1.0em;
```

```css
  width: 100%;
  background: #5318a6;
  color: #fff;
  font-weight: bold;
  border: none;
  border-radius: 6px;
  padding: 0.75em;
  font-size: 1em;
  cursor: pointer;
  transition: background 0.17s;
}
button[type="submit"]:hover, button[type="submit"]:focus {
  background: #7c3aed;
}
small.error {
  color: #d3202f;
  height: 1.2em;
  margin-top: 0.18em;
  font-size: 0.97em;
  visibility: hidden;
}
input:invalid, select:invalid {
  border-color: #e68388;
}
input.valid, select.valid {
  border-color: #39b369;
}
input.invalid, select.invalid {
  border-color: #d3202f;
}
#result {
  min-height: 1em;
  margin-top: 1.4em;
  text-align: center;
  font-weight: 600;
  color: #2b6d42;
}
@media (max-width: 600px) {
  main {
    padding: 1.2em 0.6em;
    max-width: 98vw;
  }
```

```javascript
const form = document.getElementById('regForm');
const fields = ['name', 'email', 'phone', 'dept', 'event'];
const result = document.getElementById('result');

// Validation logic
function validateField(field) {
  const input = document.getElementById(field);
  const error = input.nextElementSibling;

  let valid = true;
  let message = '';

  if (!input.value.trim()) {
    valid = false;
    message = 'This field is required';
  } else {
    switch (field) {
      case 'email':
        // Simple email regex
        const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
        valid = emailPattern.test(input.value.trim());
        message = valid ? '' : 'Enter a valid email address';
        break;
      case 'phone':
        const phone = input.value.trim();
        valid = /^\d{10}$/.test(phone);
        message = valid ? '' : 'Enter a 10-digit phone number';
        break;
    }
  }
  input.classList.toggle('invalid', !valid);
  input.classList.toggle('valid', valid);

  error.textContent = message;
  error.style.visibility = message ? 'visible' : 'hidden';
  return valid;
}

fields.forEach(field => {
  const input = document.getElementById(field);
  input.addEventListener('input', () => validateField(field));
  input.addEventListener('blur', () => validateField(field));
```

```
  input.addEventListener('blur', () => validateField(field));
});

form.addEventListener('submit', function(e) {
  e.preventDefault();
  let allValid = true;
  fields.forEach(f => {
    const valid = validateField(f);
    allValid = allValid && valid;
  });
  if (allValid) {
    result.textContent = "Registration successful!";
    result.style.color = "#2b6d42";
    form.reset();
    fields.forEach(f => {
      const input = document.getElementById(f);
      input.classList.remove('valid');
      input.classList.remove('invalid');
      input.nextElementSibling.textContent = "";
      input.nextElementSibling.style.visibility = 'hidden';
    });
  } else {
    result.textContent = "Please correct errors above.";
    result.style.color = "#d3202f";
  }
});
```



Task 4: AI-Assisted E-Commerce Product Page

Scenario:A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks : Use Copilot to generate a grid-based product catalog using HTML and CSS.

Implement an “Add to Cart” feature using JavaScript.

Modify AI-generated code to include:

    - Cart item counter at the top-right corner
    - Visual feedback when a product is added

Expected Outcome

- Functional product catalog
- Working cart counter
- Interactive user experience

CODE:

index.html styles.css script.js

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>E-Commerce Product Catalog</title>
    <link rel="stylesheet" href="styles.css">
</head>
<body>
    <header>
        <div class="header-container">
            <h1>MyStore</h1>
            <div class="cart-icon">
                <span class="cart-icon-svg">🛒</span>
                <span class="cart-counter" id="cartCounter">0</span>
            </div>
        </div>
    </header>

    <main>
        <section class="products-section">
            <h2>Featured Products</h2>
            <div class="products-grid" id="productsGrid">
                <!-- Products will be inserted here by JavaScript -->
            </div>
        </section>
    </main>

    <div class="toast-notification" id="toastNotification"></div>

    <script src="script.js"></script>
</body>
</html>
```

index.html styles.css script.js

```css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background-color: #f5f5f5;
    color: #333;
}

header {
    background-color: #2c3e50;
    color: white;
    padding: 20px 0;
    box-shadow: 0 2px 10px rgba(0, 0, 0, 0.1);
    position: sticky;
    top: 0;
    z-index: 100;
}

.header-container {
    max-width: 1200px;
    margin: 0 auto;
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 0 20px;
}

header h1 {
    font-size: 28px;
    font-weight: 600;
}

.cart-icon {
    position: relative;
    cursor: pointer;
```

index.html styles.css script.js

```css

.cart-icon {
    position: relative;
    cursor: pointer;
    font-size: 32px;
    transition: transform 0.3s ease;
}

.cart-icon:hover {
    transform: scale(1.1);
}

.cart-counter {
    position: absolute;
    top: -8px;
    right: -12px;
    background-color: #e74c3c;
    color: white;
    border-radius: 50%;
    width: 24px;
    height: 24px;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 12px;
    font-weight: bold;
    box-shadow: 0 2px 5px rgba(0, 0, 0, 0.2);
}

main {
    max-width: 1200px;
    margin: 0 auto;
    padding: 40px 20px;
}

.products-section h2 {
    margin-bottom: 30px;
    font-size: 32px;
    text-align: center;
```

index.html styles.css script.js

```css
    color: #2c3e50;
}

.products-grid {
    display: grid;
    grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
    gap: 30px;
    margin-bottom: 40px;
}

.product-card {
    background: white;
    border-radius: 8px;
    overflow: hidden;
    box-shadow: 0 2px 15px rgba(0, 0, 0, 0.1);
    transition: transform 0.3s ease, box-shadow 0.3s ease;
    cursor: pointer;
}

.product-card:hover {
    transform: translateY(-8px);
    box-shadow: 0 8px 25px rgba(0, 0, 0, 0.15);
}

.product-image {
    width: 100%;
    height: 250px;
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 80px;
    color: white;
    overflow: hidden;
}

.product-card:nth-child(2) .product-image {
    background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%);
}
```

index.html styles.css script.js

```css
}

.product-card:nth-child(3) .product-image {
    background: linear-gradient(135deg, #4facfe 0%, #00f2fe 100%);
}

.product-card:nth-child(4) .product-image {
    background: linear-gradient(135deg, #43e97b 0%, #38f9d7 100%);
}

.product-card:nth-child(5) .product-image {
    background: linear-gradient(135deg, #fa709a 0%, #fee140 100%);
}

.product-card:nth-child(6) .product-image {
    background: linear-gradient(135deg, #30cfd0 0%, #330867 100%);
}

.product-info {
    padding: 20px;
}

.product-name {
    font-size: 18px;
    font-weight: 600;
    margin-bottom: 8px;
    color: #2c3e50;
}

.product-description {
    font-size: 14px;
    color: #7f8c8d;
    margin-bottom: 15px;
    line-height: 1.4;
}

.product-footer {
    display: flex;
    justify-content: space-between;
    align-items: center;
```

index.html styles.css script.js

```css
}

.product-price {
    font-size: 22px;
    font-weight: bold;
    color: #e74c3c;
}

.add-to-cart-btn {
    background-color: #3498db;
    color: white;
    border: none;
    padding: 10px 20px;
    border-radius: 5px;
    cursor: pointer;
    font-size: 14px;
    font-weight: 600;
    transition: background-color 0.3s ease, transform 0.2s ease;
}

.add-to-cart-btn:hover {
    background-color: #2980b9;
    transform: scale(1.05);
}

.add-to-cart-btn:active {
    transform: scale(0.98);
}

.add-to-cart-btn.added {
    background-color: #27ae60;
}

/* Toast Notification Styles */
.toast-notification {
    position: fixed;
    bottom: 20px;
    right: 20px;
    background-color: #27ae60;
    color: white;
```

index.html styles.css script.js

```
    padding: 16px 24px;
    border-radius: 8px;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
    display: none;
    z-index: 1000;
    animation: slideIn 0.3s ease forwards;
    font-weight: 500;
}

.toast-notification.show {
    display: block;
}

.toast-notification.hide {
    animation: slideOut 0.3s ease forwards;
}

@keyframes slideIn {
    from {
        transform: translateX(400px);
        opacity: 0;
    }
    to {
        transform: translateX(0);
        opacity: 1;
    }
}

@keyframes slideOut {
    from {
        transform: translateX(0);
        opacity: 1;
    }
    to {
        transform: translateX(400px);
        opacity: 0;
    }
}

```

ndex.html styles.css script.js

```
/* Responsive Design */
@media (max-width: 768px) {
    .products-grid {
        grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
        gap: 20px;
    }

    header h1 {
        font-size: 22px;
    }

    .cart-icon {
        font-size: 28px;
    }

    .products-section h2 {
        font-size: 24px;
    }
}

@media (max-width: 480px) {
    .products-grid {
        grid-template-columns: repeat(auto-fill, minmax(150px, 1fr));
        gap: 15px;
    }

    main {
        padding: 20px 10px;
    }

    .product-info {
        padding: 15px;
    }

    .product-name {
        font-size: 16px;
    }

    .add-to-cart-btn {
```

index.html styles.css **script.js**

```javascript
// Product data
const products = [
    {
        id: 1,
        name: 'Wireless Headphones',
        description: 'High-quality sound with noise cancellation',
        price: '$79.99',
        emoji: '🎧'
    },
    {
        id: 2,
        name: 'Smart Watch',
        description: 'Track your fitness and stay connected',
        price: '$199.99',
        emoji: '⌚'
    },
    {
        id: 3,
        name: 'Portable Speaker',
        description: 'Powerful audio in a compact design',
        price: '$49.99',
        emoji: '🔊'
    },
    {
        id: 4,
        name: 'USB-C Cable',
        description: 'Fast charging and data transfer',
        price: '$12.99',
        emoji: '🔌'
    },
    {
        id: 5,
        name: 'Phone Case',
        description: 'Durable protection with style',
        price: '$24.99',
        emoji: '📱'
    },
    {
```

ndex.html styles.css **script.js**

```javascript
        id: 6,
        name: 'Screen Protector',
        description: 'Crystal clear protection for your screen',
        price: '$9.99',
        emoji: '🛡'
    }
];

// Cart state
let cartItems = [];

// Initialize the page
document.addEventListener('DOMContentLoaded', () => {
    renderProducts();
    loadCartFromLocalStorage();
});

// Render products in the grid
function renderProducts() {
    const productsGrid = document.getElementById('productsGrid');

    products.forEach(product => {
        const productCard = createProductCard(product);
        productsGrid.appendChild(productCard);
    });
}

// Create a product card element
function createProductCard(product) {
    const card = document.createElement('div');
    card.className = 'product-card';

    card.innerHTML = `
        <div class="product-image">
            ${product.emoji}
        </div>
        <div class="product-info">
            <div class="product-name">${product.name}</div>
            <div class="product-description">${product.description}</div>
```

x.html styles.css script.js

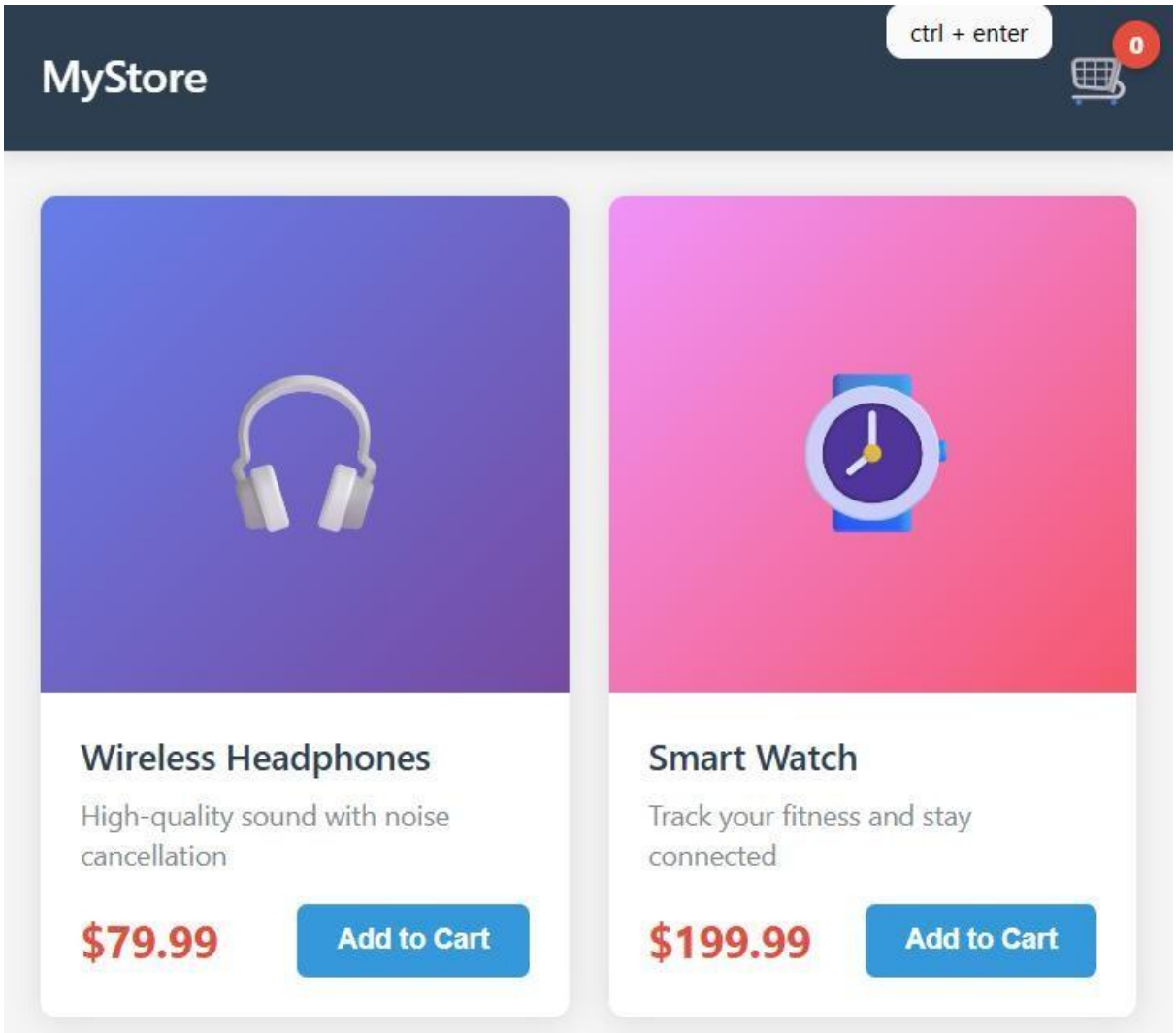
```
                <div class="product-footer">
                    <span class="product-price">${product.price}</span>
                    <button class="add-to-cart-btn" data-product-id="${product.id}">
                        Add to Cart
                    </button>
                </div>
            </div>
        `;

        // Add event listener to the button
        const button = card.querySelector('.add-to-cart-btn');
        button.addEventListener('click', (e) => {
            e.stopPropagation();
            addToCart(product, button);
        });

        return card;
    }

    // Add product to cart
    function addToCart(product, buttonElement) {
        // Add product to cart array
        cartItems.push(product);

        // Save cart to local storage
        saveCartToLocalStorage();

        // Update cart counter
        updateCartCounter();

        // Visual feedback
        showAddedFeedback(buttonElement);

        // Show toast notification
        showToastNotification(`${product.name} added to cart!`);
    }

    // Update cart counter display
    function updateCartCounter() {
```

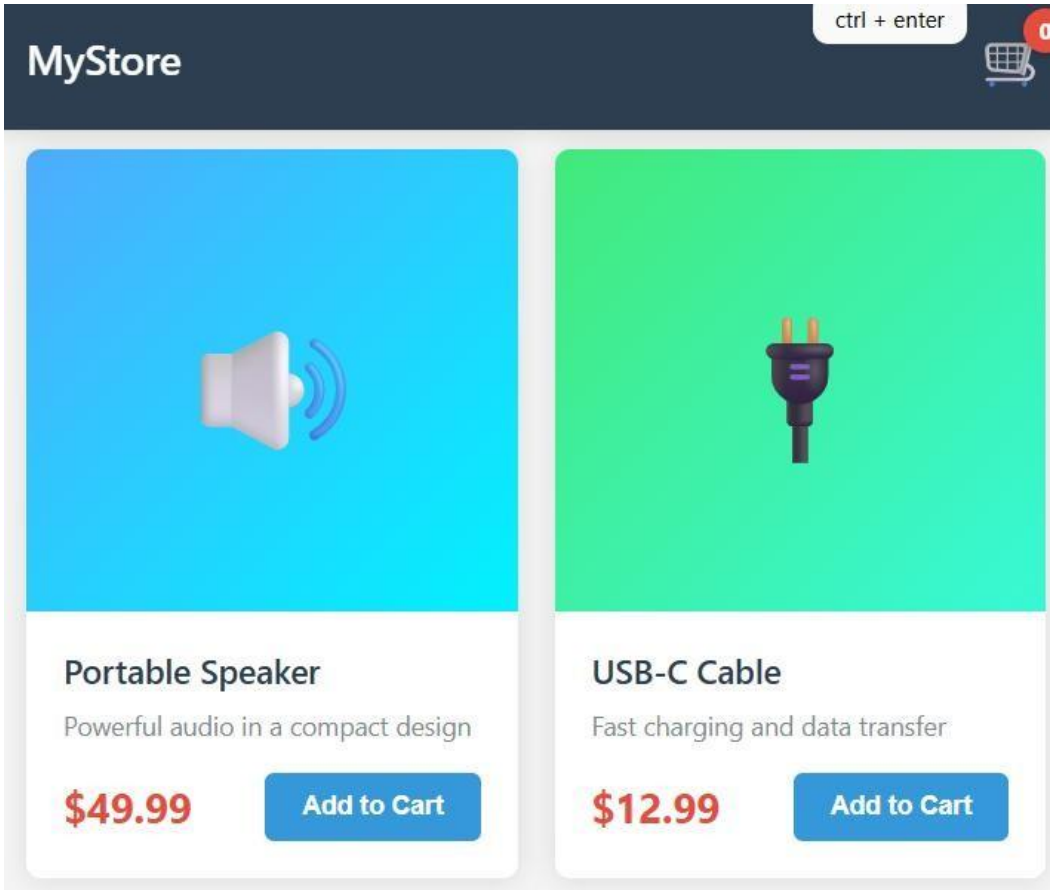
ndex.html styles.css script.js

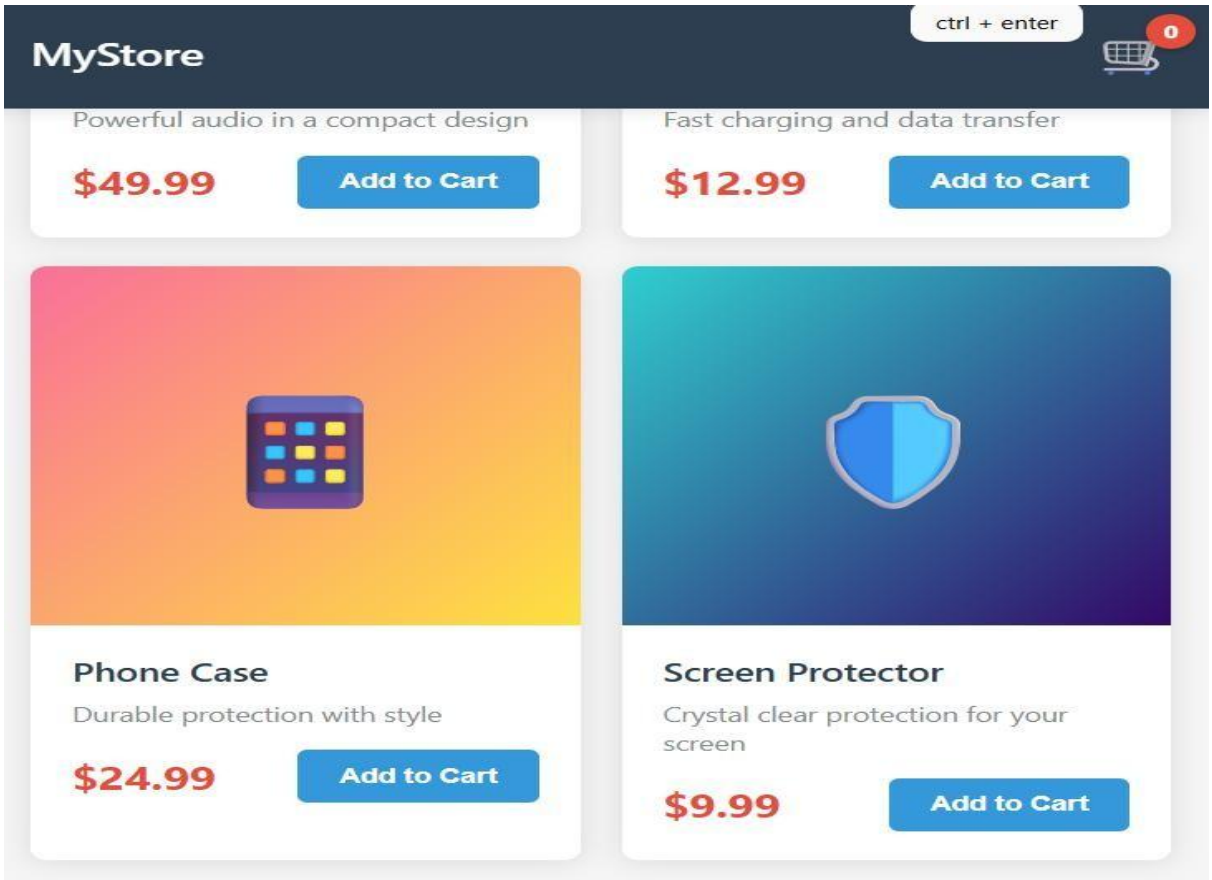
```
    const cartCounter = document.getElementById('cartCounter');
    cartCounter.textContent = cartItems.length;

    // Animate the counter
    cartCounter.style.transform = 'scale(1.3)';
    setTimeout(() => {
        cartCounter.style.transform = 'scale(1)';
    }, 200);
}

// Show visual feedback when product is added
function showAddedFeedback(buttonElement) {
    const originalText = buttonElement.textContent;
    const originalClass = buttonElement.className;

    // Change button appearance
    buttonElement.textContent = '✓ Added!';
    buttonElement.classList.add('added');
    buttonElement.disabled = true;

    // Reset after 2 seconds
    setTimeout(() => {
        buttonElement.textContent = originalText;
        buttonElement.className = originalClass;
        buttonElement.disabled = false;
    }, 2000);
}

// Show toast notification
function showToastNotification(message) {
    const toast = document.getElementById('toastNotification');
    toast.textContent = message;
    toast.classList.add('show');
    toast.classList.remove('hide');

    // Auto-hide after 3 seconds
    setTimeout(() => {
        toast.classList.add('hide');
        setTimeout(() => {
```

```
        setTimeout(() => {
            toast.classList.remove('show', 'hide');
        }, 300);
    }, 3000);
}

// Save cart to local storage
function saveCartToLocalStorage() {
    localStorage.setItem('cart', JSON.stringify(cartItems));
}

// Load cart from local storage
function loadCartFromLocalStorage() {
    const savedCart = localStorage.getItem('cart');
    if (savedCart) {
        cartItems = JSON.parse(savedCart);
        updateCartCounter();
    }
}

// Add animation to cart counter on update
const style = document.createElement('style');
style.textContent = `
    @keyframes counterPulse {
        0% {
            transform: scale(1);
        }
        50% {
            transform: scale(1.3);
        }
        100% {
            transform: scale(1);
        }
    }
`;
document.head.appendChild(style);
```

OUTPUT :




MyStore
ctrl + enter
Portable Speaker
Powerful audio in a compact design
$49.99
Add to Cart
USB-C Cable
Fast charging and data transfer
$12.99
Add to Cart


MyStore
ctrl + enter
0
Powerful audio in a compact design
$49.99
Add to Cart
Fast charging and data transfer
$12.99
Add to Cart
Phone Case
Durable protection with style
$24.99
Add to Cart
Screen Protector
Crystal clear protection for your screen
$9.99
Add to Cart

Task 5: Create a Responsive Web Page Layout

Instructions:

- Design a basic web page layout with a **header, main content area, and footer** using HTML and CSS.
- Use AI to assist in generating **responsive CSS** for different screen sizes.
- Ensure the layout is clean and visually organized. **Expected Output:**

- A responsive web page with:
- Header with navigation links
- Main section with placeholder text/images
- Footer with copyright or contact info
- Layout adapts correctly to desktop, tablet, and mobile screen sizes

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.

CODE:

index.html styles.css script.js MOHA

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Responsive Layout</title>
    <link rel="stylesheet" href="styles.css">
</head>
<body>
    <header>
        <div class="container">
            <div class="logo">Brand</div>
            <nav class="nav">
                <a href="#home">Home</a>
                <a href="#about">About</a>
                <a href="#services">Services</a>
                <a href="#contact">Contact</a>
            </nav>
            <div class="menu-toggle" id="menuToggle">☰</div>
        </div>
    </header>

    <main>
        <section id="home" class="hero">
            <div class="container">
                <h1>Welcome</h1>
                <p>Build amazing responsive websites</p>
            </div>
        </section>

        <section id="about" class="about">
            <div class="container">
                <h2>About Us</h2>
                <div class="grid">
                    <div class="card">
                        <img src="https://via.placeholder.com/150" alt="Feature 1">
                        <h3>Quality Design</h3>
                        <p>Beautiful and modern designs</p>
                    </div>
                    <div class="card">
                        <img src="https://via.placeholder.com/150" alt="Feature 2">
                        <h3>Fast Performance</h3>
```

index.html styles.css script.js

```html
                    <p>Optimized for speed</p>
                </div>
                <div class="card">
                    <img src="https://via.placeholder.com/150" alt="Feature 3">
                    <h3>Responsive</h3>
                    <p>Works on all devices</p>
                </div>
            </div>
        </div>
    </section>

    <section id="services" class="services">
        <div class="container">
            <h2>Services</h2>
            <div class="grid">
                <div class="service">
                    <h4>Web Design</h4>
                    <p>Creative web design solutions</p>
                </div>
                <div class="service">
                    <h4>Development</h4>
                    <p>Full stack development</p>
                </div>
                <div class="service">
                    <h4>SEO</h4>
                    <p>Search engine optimization</p>
                </div>
                <div class="service">
                    <h4>Support</h4>
                    <p>24/7 customer support</p>
                </div>
            </div>
        </div>
    </section>

    <section id="contact" class="contact">
        <div class="container">
            <h2>Contact Us</h2>
            <form>
                <input type="text" placeholder="Name" required>
                <input type="email" placeholder="Email" required>
                <textarea placeholder="Message" rows="5" required></textarea>
```

```html
                <button type="submit">Send</button>
            </form>
        </div>
    </section>
</main>

<footer>
    <div class="container">
        <p>&copy; 2026 Brand. All rights reserved.</p>
        <div class="footer-links">
            <a href="#home">Home</a>
            <a href="#about">About</a>
            <a href="#services">Services</a>
            <a href="#contact">Contact</a>
        </div>
    </div>
</footer>

<script src="script.js"></script>
</body>
</html>
```

index.html styles.css script.js

```css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    line-height: 1.6;
    color: #333;
}

.container {
    max-width: 1200px;
    margin: 0 auto;
    padding: 0 20px;
}

header {
    background: #2c3e50;
    color: white;
    padding: 1rem 0;
    position: sticky;
    top: 0;
    z-index: 100;
    box-shadow: 0 2px 5px rgba(0,0,0,0.1);
}

header .container {
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.logo {
    font-size: 24px;
    font-weight: bold;
    color: #3498db;
```

```css
}

.nav {
    display: flex;
    gap: 30px;
}

.nav a {
    color: white;
    text-decoration: none;
    transition: color 0.3s;
}

.nav a:hover {
    color: #3498db;
}

.menu-toggle {
    display: none;
    font-size: 24px;
    cursor: pointer;
}

.hero {
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    color: white;
    padding: 100px 20px;
    text-align: center;
}

.hero h1 {
    font-size: 48px;
    margin-bottom: 10px;
}

.hero p {
    font-size: 20px;
}

.about, .services, .contact {
    padding: 60px 20px;
}
```

```
.about {
    background: #f8f9fa;
}

section h2 {
    font-size: 36px;
    text-align: center;
    margin-bottom: 40px;
    color: #2c3e50;
}

.grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 30px;
}

.card {
    background: white;
    padding: 30px;
    border-radius: 8px;
    box-shadow: 0 2px 10px rgba(0,0,0,0.1);
    text-align: center;
    transition: transform 0.3s;
}

.card:hover {
    transform: translateY(-5px);
}

.card img {
    width: 100%;
    height: auto;
    border-radius: 5px;
    margin-bottom: 15px;
}

.card h3 {
    margin-bottom: 10px;
    color: #2c3e50;
}
```

index.html styles.css script.js

```
.card p {
    color: #7f8c8d;
}

.service {
    background: #ecf0f1;
    padding: 25px;
    border-radius: 5px;
    border-left: 4px solid #3498db;
}

.service h4 {
    margin-bottom: 10px;
    color: #2c3e50;
}

.service p {
    color: #7f8c8d;
}

.contact {
    background: #f8f9fa;
}

form {
    max-width: 500px;
    margin: 0 auto;
    display: flex;
    flex-direction: column;
    gap: 15px;
}

input, textarea {
    padding: 12px;
    border: 1px solid #ddd;
    border-radius: 5px;
    font-family: inherit;
    font-size: 14px;
}

input:focus, textarea:focus {
    outline: none;
```

index.html styles.css script.js

```css
    border-color: #3498db;
    box-shadow: 0 0 5px rgba(52, 152, 219, 0.3);
}

button {
    padding: 12px;
    background: #3498db;
    color: white;
    border: none;
    border-radius: 5px;
    font-size: 16px;
    cursor: pointer;
    transition: background 0.3s;
}

button:hover {
    background: #2980b9;
}

footer {
    background: #2c3e50;
    color: white;
    padding: 30px 20px;
    text-align: center;
}

footer .container {
    display: flex;
    justify-content: space-between;
    align-items: center;
    flex-wrap: wrap;
    gap: 20px;
}

.footer-links {
    display: flex;
    gap: 20px;
}

.footer-links a {
    color: #bdc3c7;
    text-decoration: none;
```

index.html styles.css script.js

```css
    transition: color 0.3s;
}

.footer-links a:hover {
    color: #3498db;
}

@media (max-width: 768px) {
    .menu-toggle {
        display: block;
    }

    .nav {
        position: absolute;
        top: 60px;
        left: 0;
        width: 100%;
        background: #2c3e50;
        flex-direction: column;
        gap: 0;
        padding: 20px;
        max-height: 0;
        overflow: hidden;
        transition: max-height 0.3s;
    }

    .nav.active {
        max-height: 300px;
    }

    .nav a {
        padding: 10px 0;
        border-bottom: 1px solid #34495e;
    }

    .hero h1 {
        font-size: 32px;
    }

    .hero p {
        font-size: 16px;
    }
```

index.html styles.css script.js

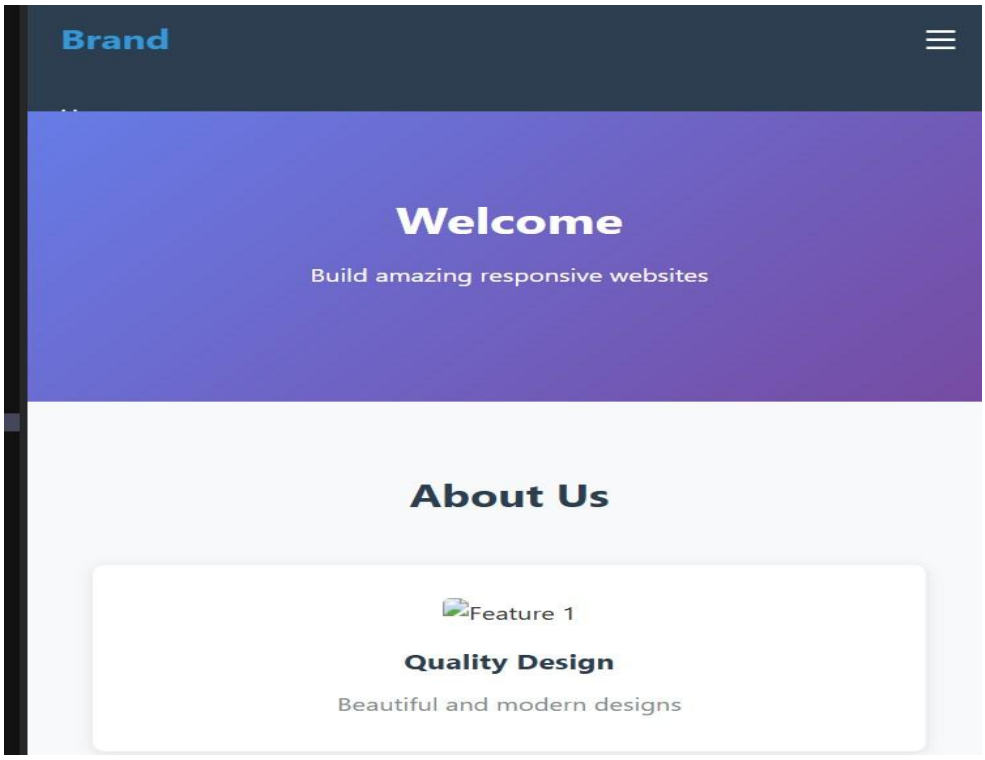
```css

    section h2 {
        font-size: 28px;
    }

    .grid {
        grid-template-columns: 1fr;
    }

    footer .container {
        flex-direction: column;
    }
}

@media (max-width: 480px) {
    .logo {
        font-size: 20px;
    }

    .hero {
        padding: 50px 20px;
    }

    .hero h1 {
        font-size: 24px;
    }

    .hero p {
        font-size: 14px;
    }

    .about, .services, .contact {
        padding: 40px 20px;
    }

    section h2 {
        font-size: 22px;
        margin-bottom: 25px;
    }

    .card, .service {
        padding: 20px;
```

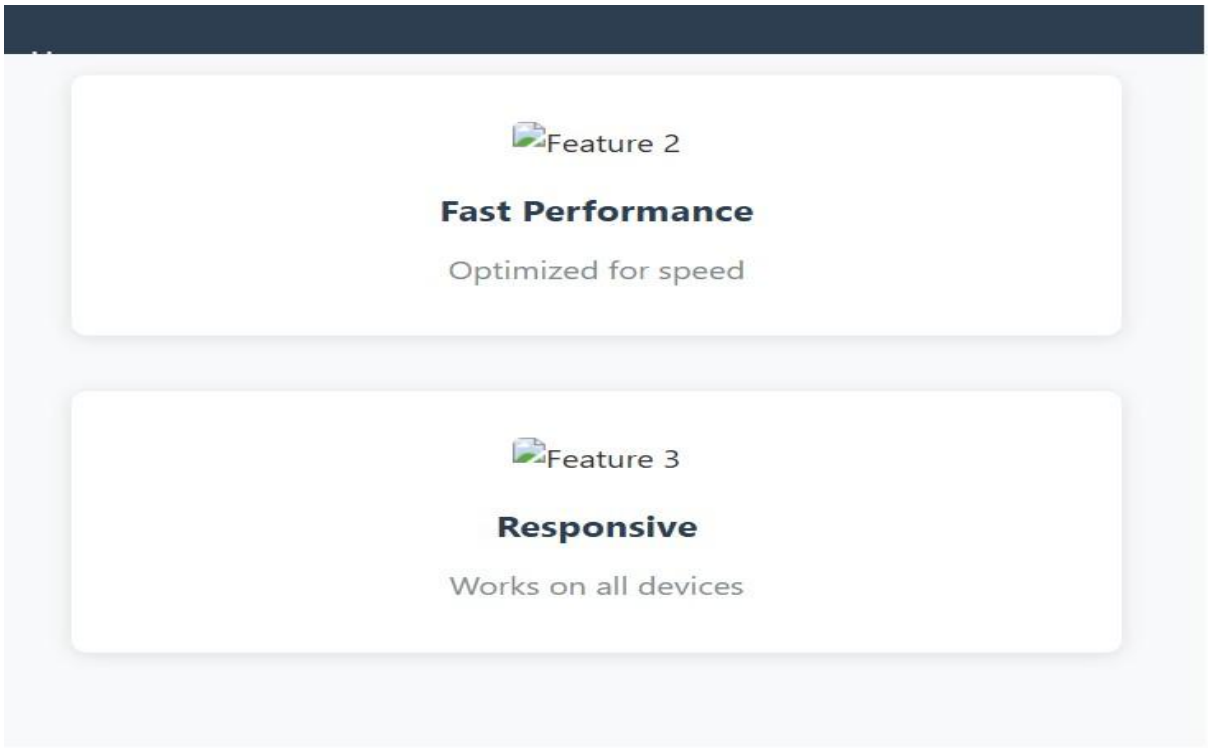
```css
    .card, .service {
        padding: 20px;
    }

    .footer-links {
        flex-direction: column;
        gap: 10px;
    }
}
```

index.html styles.css script.js

```javascript
const hamburger = document.getElementById('hamburger');
const navbar = document.querySelector('.navbar');

hamburger.addEventListener('click', () => {
    hamburger.classList.toggle('active');
    navbar.classList.toggle('active');
});

document.querySelectorAll('.nav-link').forEach(link => {
    link.addEventListener('click', () => {
        hamburger.classList.remove('active');
        navbar.classList.remove('active');
    });
});

document.querySelector('.contact-form').addEventListener('submit', (e) => {
    e.preventDefault();
    alert('Thank you for your message!');
    e.target.reset();
});

document.querySelector('.cta-btn').addEventListener('click', () => {
    document.getElementById('services').scrollIntoView({behavior: 'smooth'});
});
```

OUTPUT :


Brand
Welcome
Build amazing responsive websites
About Us
Feature 1
Quality Design
Beautiful and modern designs


Feature 2
Fast Performance
Optimized for speed
Feature 3
Responsive
Works on all devices

Services

Brand

Web Design

Creative web design solutions

Development

Full stack development

SEO

Search engine optimization

Support

24/7 customer support

Contact Us

Name

Email

Message

Send